

Chapter 3.5 – Induction Bias

Detailed classroom presentation: Induction Bias, Bias, Variance and Best Fit

Machine Learning – Basics of Learning Theory

Section 3.5 of the prescribed textbook

Subsections: 3.5 Induction Bias; 3.5.1 Bias and Variance; 3.5.2 Bias vs Variance Tradeoff; 3.5.3 Best Fit in Machine Learning

Includes conceptual explanations, comparison tables, classroom examples and review questions

Learning Objectives

Define induction and induction bias in the context of machine learning.

Explain the two forms of induction bias: constraint/restriction and preference.

Explain prediction error, bias error and variance error.

Interpret the effect of high/low bias and high/low variance on learning models.

Explain the bias–variance trade-off and its importance in model selection.

Explain overfitting, underfitting and the idea of a best-fit model.

Describe techniques discussed in the textbook for reducing variance: regularization, early stopping/pruning and ensemble learning.

3.5 Induction Bias – Concept

Induction is a process of learning a target function or generalizing training data into a general model.

Induction uses assumptions considered by the learning algorithm to perform learning from the available examples.

These assumptions are called the induction bias of the algorithm.

Induction bias is also described as the bias of the algorithm.

Because the hypothesis space can contain many possible hypotheses, the learner needs assumptions or preferences to guide the search.

Why Does a Learning Algorithm Need Bias?

Training examples alone may not uniquely determine one correct hypothesis.

A learner therefore needs assumptions to decide which hypotheses are plausible or preferable.

Bias can restrict the set of hypotheses considered.

Bias can also impose an ordering or preference over hypotheses.

Thus, learning is not merely memorizing training data; it involves selecting a useful generalization.

Two Types of Induction Bias

Type	Meaning	Purpose
Constraint / Restriction	Restricts the hypothesis space or imposes conditions on allowable hypotheses.	Reduces the set of hypotheses that the learner must consider.
Preference	Imposes an ordering or preference on the hypothesis space.	Guides the learner toward preferred hypotheses during search.

Examples of Algorithmic Assumptions

The textbook illustrates that different learning algorithms use different assumptions.

Decision-tree learning: assumes a preference for particular ways of selecting attributes/tests.

k-Nearest Neighbour: relies on the idea that nearby instances tend to be similar.

Naive Bayes: assumes input variables are independent.

These assumptions influence how the algorithm searches for or represents the learned model.

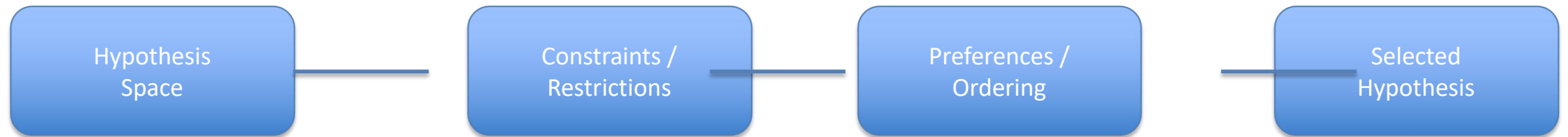
Induction Bias and Hypothesis Space

Hypothesis space = the set of possible hypotheses considered by a learner.

Constraint bias reduces the hypotheses that are eligible for consideration.

Preference bias does not necessarily remove hypotheses; it can determine which hypotheses are preferred or considered first.

Learning can therefore be viewed as searching a hypothesis space under the assumptions of the learning algorithm.



3.5.1 Bias and Variance

For an input x , the model produces a predicted output $\hat{y}$.

If $\hat{y}$ differs from the actual output y , the prediction contains error.

The textbook discusses three kinds of prediction error:

- Irreducible error – associated with unknown factors, noise and other effects that cannot be eliminated.

- Bias error – associated with the difference between the predicted output and the actual output of the learning model.

- Variance error – associated with changes in the target function when the training data changes.

Irreducible Error

Irreducible error is caused by factors that are unknown or inherently noisy.

It cannot be completely removed simply by changing the learning algorithm.

Examples include measurement noise or information not available to the model.

Important distinction: reducing model complexity or changing the hypothesis does not eliminate information that is absent or noisy in the data.

Bias Error – Detailed Interpretation

Bias reflects the effect of simplifying assumptions made by a learning model.

High bias indicates that the model makes strong simplifying assumptions.

A high-bias model may systematically miss important relationships in the data.

Bias is therefore related to the model's ability to represent the underlying target relationship.

Classroom interpretation: if a model is consistently too simple, its predictions can remain systematically inaccurate even when more training examples are supplied.

Variance Error – Detailed Interpretation

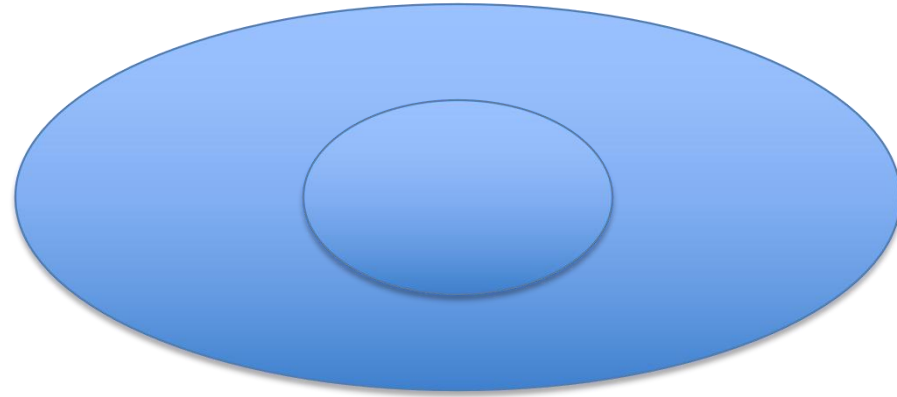
Variance describes how much the learned target function can change when the training data changes.

A high-variance model is sensitive to the particular training sample used.

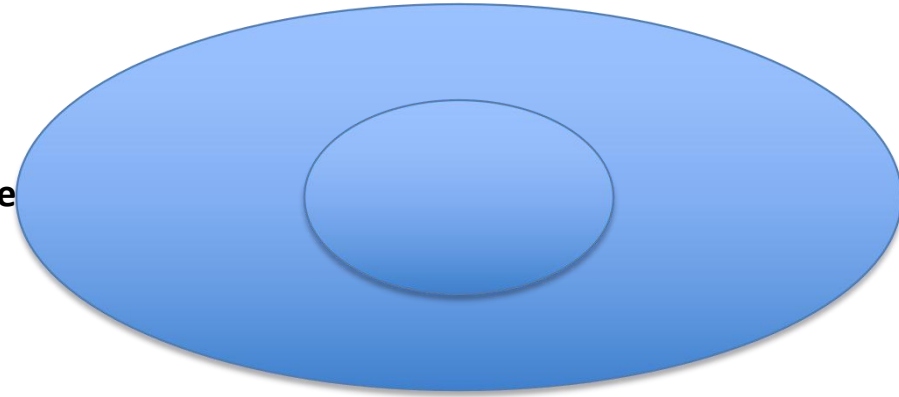
Such a model may fit the training data closely but may not perform consistently on new data.

Variance is therefore closely related to model sensitivity and generalization.

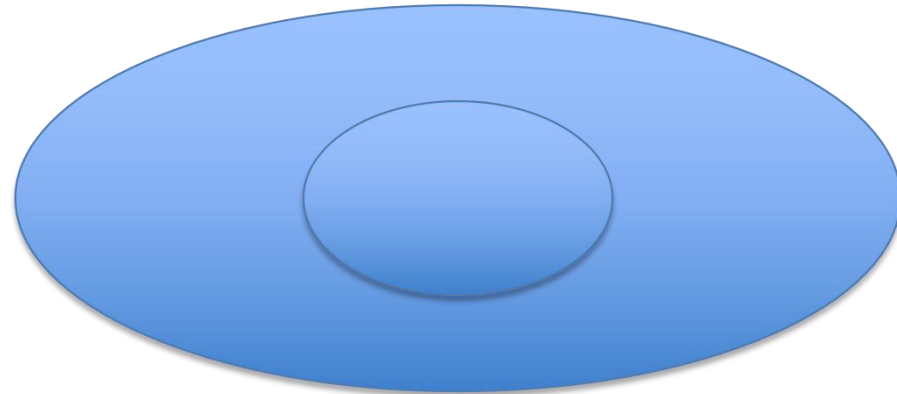
Bias vs Variance – Pictorial Interpretation



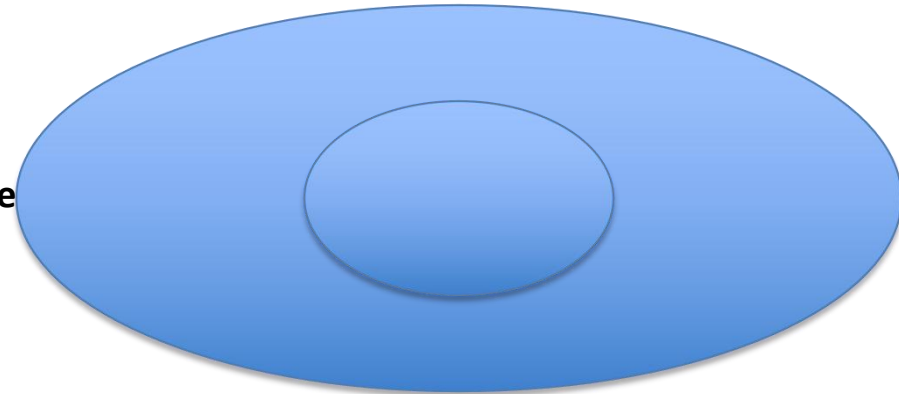
**High Bias
Low Variance**



**High Bias
High Variance**



**Low Bias
Low Variance**



**Low Bias
High Variance**

Reading the Four Bias–Variance Cases

Cas e	Bias	Varianc e	Typical interpretation
1	High	Low	Model is consistently simplified; predictions may be systematically away from target.
2	High	High	Model has strong simplifying assumptions but is also sensitive to training-data changes.
3	Low	Low	Desired stable learning behaviour; model captures useful structure without excessive sensitivity.
4	Low	High	Model can fit complex patterns but may change substantially with the training sample.

3.5.2 Bias vs Variance Trade-off

There is a trade-off between bias and variance.

Changing model assumptions and complexity can alter both sources of error.

Low-bias models generally make fewer simplifying assumptions but can become more sensitive to the training data.

High-bias models generally make more simplifying assumptions and can be more stable, but may fail to represent the target relationship adequately.

The practical objective is to balance these effects so that the learned model generalizes well.

Textbook Trade-off Table – Bias

Bias level	Textbook interpretation	Examples given in the textbook
Low bias	Less simplifying assumptions.	Decision Trees, k-Nearest Neighbours (k-NN), Support Vector Machines (SVM).
High bias	More simplifying assumptions or considerations that may restrict the target function.	Linear Regression, Linear Discriminant Analysis (LDA), Logistic Regression, Naive Bayes.

Textbook Trade-off Table – Variance

Variance level	Textbook interpretation	Examples given in the textbook
Low variance	Training data is more general; low-variance algorithms show only slight differences when training data changes.	Linear Regression, LDA, Logistic Regression.
High variance	Training data is more specific; predictions can differ more when the training data changes.	Decision Trees, k-NN, SVM.

Important Observation: Model Complexity

Linear machine-learning algorithms are described in the textbook as often exhibiting high bias and low variance.

Non-linear machine-learning algorithms are described as exhibiting low bias and high variance.

Therefore, increasing model flexibility is not automatically better.

A more flexible model may capture complex relationships but can become sensitive to the training sample.

A simpler model may be stable but can fail to capture the required relationship.

Overfitting and the Bias–Variance Trade-off

The textbook associates low bias and high variance with the possibility of overfitting.

An overfitted model can perform very well on training data but poorly on unseen data.

Regularization can reduce the effect of excessive model complexity.

Adding more training instances, reducing the feature set, or reducing model complexity can help address high variance, as discussed in the text.

Goal: improve generalization rather than maximize performance only on the training set.

Regularization

Regularization is presented as one way of reducing variance.

It reduces the number/effect of parameters during the training phase for parametric learning algorithms.

The purpose is to control model complexity and reduce the tendency to fit training data too closely.

Teaching point: regularization trades some flexibility for improved generalization.

Early Stopping and Pruning

For tree-based algorithms, early stopping or pruning can help reduce variance.

Pruning removes or prevents unnecessarily complex parts of a decision tree.

Early stopping prevents the tree from continuing to grow when further growth may increase sensitivity to training data.

These techniques control model complexity and can improve generalization.

Ensemble Learning and Variance

Ensemble learning is another technique mentioned for reducing variance.

The idea is to train models on multiple datasets and average their predictions.

Combining predictions can make the final prediction more stable than relying on one learned model.

The textbook connects ensemble learning with improving generalization and reducing variance.

Illustrative Example – Choosing a Model

Suppose a dataset contains examples of students classified as likely to receive a job offer.

Model A: a very simple rule uses only one attribute.

Advantage: simple and stable; risk: may ignore important patterns → high bias.

Model B: a very complex model fits almost every training instance.

Advantage: captures detailed patterns; risk: changes considerably with training data → high variance.

Model C: a controlled model captures the important patterns while avoiding unnecessary complexity.

Goal: better generalization to new students → a practical best-fit choice.

3.5.3 Best Fit in Machine Learning

A hypothesis obtained from training instances for a specific model is called inductive learning.

Generalization is the ability of the model to correctly predict new data after training.

If a model is over-trained or under-trained, its performance can deteriorate.

The best-fit model should provide good performance on unseen data.

The textbook emphasizes the balance between underfitting and overfitting.

Underfitting – Concept

Underfitting occurs when the model is too simple or has not learned enough from the training data.

The model fails to capture important patterns in the training data.

Typical indication: poor predictive performance because the model is unable to represent the target relationship adequately.

Connection with Section 3.5: excessive bias is a major concern when the model is too restrictive or simplistic.

Overfitting – Concept

Overfitting occurs when a model learns too much from the training instances.

The model can fit training examples very closely but may fail to generalize.

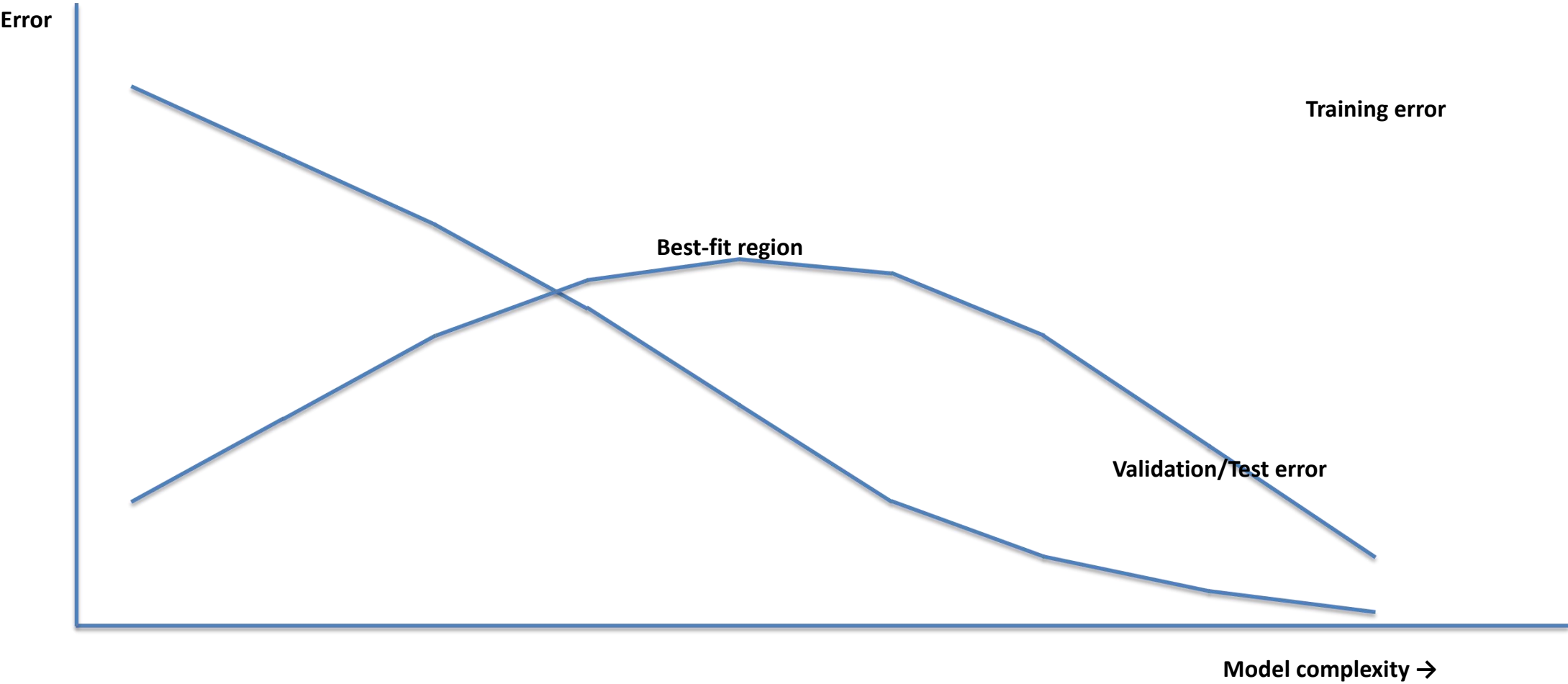
The textbook describes overfitting as a condition in which the model's performance on unseen/test data can deteriorate.

High variance is closely associated with this behaviour.

Underfitting vs Overfitting vs Best Fit

Aspect	Underfitting	Best Fit	Overfitting
Model complexity	Too simple	Appropriate for the problem	Too complex
Learning behaviour	Learns too little	Learns useful structure	Learns too much from training instances
Training data	Important patterns may be missed	Patterns captured appropriately	Training instances may be fitted very closely
Unseen data	Poor generalization	Good generalization is the objective	Performance may deteriorate
Main concern	High bias / insufficient learning	Balance	High variance / excessive fitting

Best Fit – A Conceptual Learning Curve



Based on S. Sridhar & M. Vijayalakshmi, Machine Learning — Chapter 3, Illustrative teaching diagram; Section 3.5.3 provides the underfitting/best-fit/overfitting interpretation

End-to-End Decision Process

Step 1: Identify the learning problem and available training data.

Step 2: Select an appropriate hypothesis/model family.

Step 3: Consider the assumptions introduced by the algorithm — its induction bias.

Step 4: Evaluate whether the model is too simple or too complex.

Step 5: Examine generalization performance on unseen data.

Step 6: If variance is high, consider regularization, pruning/early stopping, ensemble learning or other complexity-control approaches discussed in the text.

Step 7: Select the model that provides the desired balance and best generalization.

Key Takeaways

Induction bias consists of assumptions used by a learning algorithm while generalizing from training data.

Two types: constraint/restriction and preference.

Prediction error can include irreducible error, bias error and variance error.

High bias is associated with stronger simplifying assumptions.

High variance is associated with greater sensitivity to changes in training data.

Bias and variance must be considered together — this is the bias–variance trade-off.

Regularization, pruning/early stopping and ensemble learning are discussed as ways to control variance.

Best fit means achieving useful generalization rather than merely fitting the training instances.

Exam-Oriented Questions – L2 / L3

L2: Define induction bias. Explain its two types.

L2: Differentiate bias and variance.

L2: What is meant by irreducible error?

L3: Explain why a learning algorithm needs induction bias.

L3: Explain the bias–variance trade-off with suitable examples.

L3: Compare high-bias and high-variance models.

L3: Explain how regularization can reduce variance.

L3: Explain underfitting, overfitting and best fit.

Exam-Oriented Questions – L4 / Application

L4: A model performs very well on training data but poorly on unseen data. Identify the likely problem and explain the bias/variance interpretation.

L4: Compare a simple linear model with a highly flexible non-linear model in terms of bias and variance.

L4: Given two candidate models, recommend which one is preferable when the second model changes substantially with different training samples.

L4: Explain how regularization, pruning and ensemble learning can be used to improve generalization.

L4: Design a model-selection strategy that aims to avoid both underfitting and overfitting.

Quick Classroom Activity

Form groups of 3–4 students.

Give each group two hypothetical models for the same classification task.

Model A: simple rule using one feature.

Model B: highly complex rule matching almost every training example.

Ask students to identify likely bias/variance behaviour.

Ask: Which model is likely to generalize better? What additional evidence would you seek?

Discuss regularization/pruning/ensemble learning as possible remedies.

Section 3.5 – One-Slide Revision

Concept	Remember
Induction Bias	Prior assumptions used by a learning algorithm.
Constraint / Restriction	Limits the hypothesis space.
Preference	Orders or prefers hypotheses in the hypothesis space.
Bias	Error associated with simplifying assumptions.
Variance	Error/change associated with sensitivity to training-data changes.
Trade-off	Balance bias and variance for good generalization.
Variance control	Regularization; early stopping/pruning; ensemble learning.
Best Fit	Model that generalizes well without underfitting or overfitting.